

PREPARED FOR

GNSS Survey Report

40.87092, -121.34479

Section 30, T35N, R6E, Mount Diablo Meridian

✓ CORS Elev Delta: 712 ft below nearest CORS station (P348, 25.4 mi) – Good baseline geometry.

COORDINATES	ELEVATION	STATE PLANE	FEMA ZONE	DECLINATION
40.8709°N 121.3448°W WGS84	4,762 ft NAVD88 · MSL	CA 1 (0401) · N 2,201,230.04 E 6,742,858.49 (US ft, NAD83)	Zone D N/A	13.3° E WMM 2026

APN 031380001000 | ZONING U | ACRES 613.110 ac

BEST GNSS WINDOW Friday Jul 24 · 12:00 PM – 4:00 PM | PDOP 0.91 · 34 sv | NEAREST CORS
P348 · 25.4 mi ✓

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VIEW FIELD MAP ↗

SITE SATELLITE ↗

EMERGENCY MAP ↗

SCAN FOR KMZ
Site file · Google Earth
↓ Download KMZ

ASSESSOR-RECORDER
Shasta County
Assessor-Recorder
· 1450 Court St.
Redding, CA 96004
NEAREST BM
LT0642 · 2.7 mi
NEAREST CORS
P348 · 25.4 mi
GENERATED
July 22,
2026
SYSTEM
TopoScout ·
info@toposcout.com



INTERACTIVE REPORT

This report includes clickable links and QR codes throughout – tap or scan them to open maps, official data sources, and field tools directly. Your site is also available as a drafting-ready CAD (DXF) file with georeferenced aerials. Add it anytime at toposcout.com.

SITE & SIGNAL REPORT

COORDINATES

40.8709, -121.3448

WGS84 / NAD83

ELEVATION

4,762 ft

Above MSL (approx)

MAGNETIC DECLINATION

13.3° E

Compass reads 13.3° too far W

FEMA FLOOD ZONE

D

See control points page · Panel N/A

IONOSPHERIC CONDITIONS

PEAK KP INDEX

5.0

AVG KP

2.0

SOLAR FLUX F10.7

138 sfu

TEC (VTEC) ▲ DEGRADED

18.4 TECU

-2.0 TECU vs median

ASSESSMENT

ELEVATED ACTIVITY

Peak Kp 5.0 (avg 2.0). Solar flux F10.7 = 138 sfu. Elevated geomagnetic activity. Potential for significant ionospheric delays and cycle slips. Consider postponing precision work if possible.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Wed, Jul 22	0.98	4:00 PM – 8:00 PM	33	Ideal
Thu, Jul 23	0.99	3:00 PM – 7:00 PM	32	Ideal
Fri, Jul 24	0.97	12:00 PM – 4:00 PM	34	Ideal
Sat, Jul 25	0.95	3:00 PM – 7:00 PM	36	Ideal
Sun, Jul 26	0.97	1:00 PM – 5:00 PM	35	Ideal
Mon, Jul 27	0.98	4:00 PM – 8:00 PM	32	Ideal
Tue, Jul 28	0.98	1:00 PM – 5:00 PM	33	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.

Days analyzed: 7 Recommended work days: 5 of 7 Best PDOP: 0.95

►

RECOMMENDED WINDOW

Tue, Jul 28

6:00 AM – 10:00 AM

Avg wind 2 mph · Gusts 3 mph

Ideal conditions — 4-hr static occupation

ALTERNATE WINDOW

Wed, Jul 22

5:00 AM – 9:00 AM

Avg wind 3 mph · Gusts 4 mph

Ideal conditions — 4-hr static occupation

 WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 22, 2026) · Conditions may change – verify before fieldwork

 GPS CONSTELLATION HEALTH

 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

 Land Cover & Multipath Risk

SKY OBSTRUCTION

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

MULTIPATH RISK

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13°

Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.8–0.9 across survey windows

PDOP remains ≤ 0.9 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

High 84°F

Low 65°F

Wind max 15 mph

Gusts 20 mph

Precip 0.00"

Sunrise 5:52 AM

Sunset 8:30 PM

First Quarter 53%

GNSS: Avg PDOP **0.98** · Best 4-hr window 4:00 PM – 8:00 PM (PDOP 0.93, 33 sats avg)

GNSS / DOP - HOURLY			
	PDOP	SATS	RATING
6:00 AM	1.01	31 sv	GOOD
7:00 AM	0.92	32 sv	IDEAL
8:00 AM	0.90	36 sv	IDEAL
9:00 AM	1.00	31 sv	GOOD
10:00 AM	0.97	30 sv	IDEAL
11:00 AM	1.17	25 sv	GOOD
12:00 PM	0.98	28 sv	IDEAL
1:00 PM	1.06	28 sv	GOOD
2:00 PM	1.04	33 sv	GOOD
3:00 PM	0.95	35 sv	IDEAL
4:00 PM	0.90	35 sv	IDEAL
5:00 PM	0.93	33 sv	IDEAL
6:00 PM	0.94	31 sv	IDEAL
7:00 PM	0.94	32 sv	IDEAL

GO – GOOD CONDITIONS

Partly cloudy | Wind: 15.4 mph (gusts 20.4 mph) | Precip: 0" | Temp: 64.7°–83.8°F

DRONE WIND ANALYSIS

CAUTION Peak: 15.4/20.4 mph W

* Sunrise: 5:52 AM ☾ Sunset: 8:30 PM fly window clipped to daylight

☒ Best flight window: 6:00 AM – 9:00 PM (15 hrs)

	SUSTAINED / GUSTS mph	≈turbulence	DIR
5 AM	4/4 mph		E
6 AM	2/3 mph		N
7 AM	3/4 mph		WNW
8 AM	4/4 mph		WNW
9 AM	5/9 mph		NW
10 AM	5/12 mph		NNW
11 AM	6/12 mph		NW
12 PM	8/17 mph		NW
1 PM	10/17 mph		NW
2 PM	13/18 mph		NW
3 PM	13/16 mph		NW
4 PM	13/16 mph		WSW
5 PM	15/20 mph		WSW
6 PM	14/19 mph		WSW
7 PM	13/18 mph		WSW
8 PM	11/16 mph		WSW

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE **CLASS G** ▶ FREE TO FLY
Class G uncontrolled airspace - fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 89°F

Low 56°F

Wind max 13 mph

Gusts 18 mph

Precip 0.00"

Sunrise 5:53 AM

Sunset 8:30 PM

First Quarter 64%

GNSS: Avg PDOP **0.99** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.97, 32 sats avg)

GNSS / DOP - HOURLY					
	PDOP		SATS		RATING
6:00 AM	0.96		32 sv		IDEAL
7:00 AM	0.93		34 sv		IDEAL
8:00 AM	0.94		32 sv		IDEAL
9:00 AM	1.12		26 sv		GOOD
10:00 AM	0.92		31 sv		IDEAL
11:00 AM	1.09		26 sv		GOOD
12:00 PM	0.94		32 sv		IDEAL
1:00 PM	1.07		29 sv		GOOD
2:00 PM	0.97		31 sv		IDEAL
3:00 PM	0.94		34 sv		IDEAL
4:00 PM	0.98		32 sv		IDEAL
5:00 PM	0.99		32 sv		IDEAL
6:00 PM	0.95		31 sv		IDEAL
7:00 PM	1.10		26 sv		GOOD

GO – GOOD CONDITIONS

Partly cloudy | Wind: 13.3 mph (gusts 17.8 mph) | Precip: 0" | Temp: 55.8°–88.6°F

DRONE WIND ANALYSIS

GO

Peak: 13.3/17.8 mph W

* Sunrise: 5:53 AM

☾ Sunset: 8:30 PM

fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≠turbulence	DIR
5 AM	4/4 mph		E
6 AM	4/4 mph		E
7 AM	5/5 mph		ESE
8 AM	2/6 mph		ENE
9 AM	2/9 mph		N
10 AM	4/11 mph		NNW
11 AM	6/12 mph		NW
12 PM	8/13 mph		NW
1 PM	9/14 mph		WNW
2 PM	9/14 mph		WNW
3 PM	11/18 mph		WNW
4 PM	12/17 mph		WNW
5 PM	13/16 mph		W
6 PM	12/13 mph		WSW
7 PM	9/10 mph		W
8 PM	5/5 mph		WSW

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS G ▶ FREE TO FLY

Class G uncontrolled airspace - fly under 400ft AGL, follow Part 107

→ TFR STATUS NO TFRs WITHIN 30 MI [Check B4UFLY](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 93°F

Low 55°F

Wind max 14 mph

Gusts 22 mph

Precip 0.00"

Sunrise 5:54 AM

Sunset 8:29 PM

☾ Waning Gibbous 74%

GNSS: Avg PDOP **0.97** · Best 4-hr window 12:00 PM – 4:00 PM (PDOP 0.91, 34 sats avg)

GNSS / DOP - HOURLY					
	PDOP		SATS		RATING
6:00 AM	0.99		29 sv		IDEAL
7:00 AM	0.87		34 sv		IDEAL
8:00 AM	0.89		35 sv		IDEAL
9:00 AM	1.10		28 sv		GOOD
10:00 AM	1.01		30 sv		GOOD
11:00 AM	1.11		27 sv		GOOD
12:00 PM	0.90		32 sv		IDEAL
1:00 PM	0.95		33 sv		IDEAL
2:00 PM	0.91		34 sv		IDEAL
3:00 PM	0.89		37 sv		IDEAL
4:00 PM	1.03		33 sv		GOOD
5:00 PM	0.94		31 sv		IDEAL
6:00 PM	1.00		28 sv		GOOD
7:00 PM	0.94		30 sv		IDEAL



GO – GOOD CONDITIONS

Clear sky | Wind: 14.3 mph (gusts 22.1 mph) | Precip: 0" | Temp: 55.1°–92.9°F



DRONE WIND ANALYSIS



CAUTION Peak: 14.3/22.1 mph SSW

* Sunrise: 5:54 AM ☾ Sunset: 8:29 PM fly window clipped to daylight

✓ Best flight window: 6:00 AM – 2:00 PM (8 hrs)

🔄 Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤turbulence	DIR
5 AM	7/6 mph		SSE
6 AM	6/6 mph		SSE
7 AM	5/6 mph		SSE
8 AM	2/6 mph		S
9 AM	3/12 mph		WNW
10 AM	7/17 mph		WNW
11 AM	7/18 mph		NW
12 PM	8/19 mph		WNW
1 PM	9/20 mph		WSW
2 PM	14/22 mph		SW
3 PM	14/22 mph		SW
4 PM	14/22 mph		SW
5 PM	14/22 mph		WSW
6 PM	14/19 mph		WSW
7 PM	10/16 mph		WSW
8 PM	5/5 mph		SW

BEST WINDOW

6:00 AM – 2:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS G ► FREE TO FLY
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→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#) ↗
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

#4

Saturday, Jul 25

WEEKEND

Overcast

16 mph

GO

High 90°F Low 54°F Wind max 16 mph Gusts 22 mph Precip 0.00" Sunrise 5:55 AM Sunset 8:28 PM

Waxing Gibbous 82%

GNSS: Avg PDOP 0.95 · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.88, 36 sats avg)

GNSS / DOP - HOURLY

	PDOP	SATS	RATING
6:00 AM	0.96	33 sv	IDEAL
7:00 AM	1.04	31 sv	GOOD
8:00 AM	0.92	36 sv	IDEAL
9:00 AM	1.12	28 sv	GOOD
10:00 AM	0.91	31 sv	IDEAL
11:00 AM	1.04	28 sv	GOOD
12:00 PM	0.91	32 sv	IDEAL
1:00 PM	0.99	33 sv	IDEAL
2:00 PM	1.00	33 sv	GOOD
3:00 PM	0.84	36 sv	IDEAL
4:00 PM	0.89	36 sv	IDEAL
5:00 PM	0.92	35 sv	IDEAL
6:00 PM	0.85	37 sv	IDEAL
7:00 PM	0.96	31 sv	IDEAL



GO - GOOD CONDITIONS

Partly cloudy | Wind: 16.1 mph (gusts 21.5 mph) | Precip: 0" | Temp: 54.2°–89.5°F



DRONE WIND ANALYSIS



CAUTION Peak: 16.1/21.5 mph SW

* Sunrise: 5:55 AM Sunset: 8:28 PM fly window clipped to daylight

Best flight window: 6:00 AM – 11:00 AM (5 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤turbulence	DIR
5 AM	6/8 mph		SSE
6 AM	6/7 mph		SSE
7 AM	5/8 mph		SSE
8 AM	2/11 mph		SW
9 AM	4/14 mph		WNW
10 AM	7/18 mph		WNW
11 AM	8/21 mph		WNW
12 PM	10/21 mph		W
1 PM	12/21 mph		WSW
2 PM	15/22 mph		WSW
3 PM	16/21 mph		WSW
4 PM	16/21 mph		WSW
5 PM	15/21 mph		WSW
6 PM	13/18 mph		WSW
7 PM	10/15 mph		WSW
8 PM	6/7 mph		SW

BEST WINDOW

6:00 AM – 11:00 AM

Good conditions with 5-hr clean window available

AIRSPACE CLASS G ► FREE TO FLY

Class G uncontrolled airspace - fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

#5

Sunday, Jul 26

WEEKEND

Clear sky

15 mph

GO

High 90°F

Low 52°F

Wind max 15 mph

Gusts 16 mph

Precip 0.00"

Sunrise 5:56 AM

Sunset 8:27 PM

Waxing Gibbous 90%

GNSS: Avg PDOP 0.97 · Best 4-hr window 1:00 PM – 5:00 PM (PDOP 0.92, 35 sats avg)

GNSS / DOP - HOURLY

	PDOP	SATS	RATING
6:00 AM	0.90	34 sv	IDEAL
7:00 AM	1.09	28 sv	GOOD
8:00 AM	0.92	32 sv	IDEAL
9:00 AM	1.08	28 sv	GOOD
10:00 AM	1.01	29 sv	GOOD
11:00 AM	0.96	28 sv	IDEAL
12:00 PM	0.97	28 sv	IDEAL
1:00 PM	0.92	33 sv	IDEAL
2:00 PM	0.95	31 sv	IDEAL
3:00 PM	0.89	38 sv	IDEAL
4:00 PM	0.91	36 sv	IDEAL
5:00 PM	1.18	28 sv	GOOD
6:00 PM	0.95	31 sv	IDEAL
7:00 PM	0.92	31 sv	IDEAL



GO - GOOD CONDITIONS

Clear sky | Wind: 14.7 mph (gusts 16.1 mph) | Precip: 0" | Temp: 52.4°–89.6°F



DRONE WIND ANALYSIS

GO Peak: 14.7/16.1 mph WSW

* Sunrise: 5:56 AM Sunset: 8:27 PM fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≠turbulence	DIR
5 AM	6/5 mph		SSE
6 AM	6/5 mph		SSE
7 AM	5/4 mph		SSE
8 AM	1/2 mph		SSE
9 AM	3/5 mph		NNW
10 AM	6/9 mph		NW
11 AM	8/12 mph		NNW
12 PM	9/14 mph		NW
1 PM	11/15 mph		NW
2 PM	12/16 mph		WNW
3 PM	13/16 mph		WNW
4 PM	14/15 mph		W
5 PM	15/14 mph		W
6 PM	13/12 mph		WSW
7 PM	9/9 mph		WSW
8 PM	6/6 mph		WSW

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS G ► FREE TO FLY
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→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#)
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 91°F

Low 54°F

Wind max 15 mph

Gusts 15 mph

Precip 0.00"

Sunrise 5:57 AM

Sunset 8:26 PM

☾ Waxing Gibbous 95%

GNSS: Avg PDOP **0.98** · Best 4-hr window 4:00 PM – 8:00 PM (PDOP 0.95, 32 sats avg)

GNSS / DOP - HOURLY					
	PDOP		SATS		RATING
6:00 AM	0.91		34 sv		IDEAL
7:00 AM	0.94		33 sv		IDEAL
8:00 AM	0.95		36 sv		IDEAL
9:00 AM	0.99		31 sv		IDEAL
10:00 AM	1.08		26 sv		GOOD
11:00 AM	0.99		27 sv		IDEAL
12:00 PM	1.01		27 sv		GOOD
1:00 PM	0.95		33 sv		IDEAL
2:00 PM	0.99		33 sv		IDEAL
3:00 PM	1.11		29 sv		GOOD
4:00 PM	0.91		35 sv		IDEAL
5:00 PM	1.01		29 sv		GOOD
6:00 PM	0.97		31 sv		IDEAL
7:00 PM	0.90		32 sv		IDEAL



GO – GOOD CONDITIONS

Partly cloudy | Wind: 14.6 mph (gusts 15 mph) | Precip: 0" | Temp: 54.3°–90.8°F

 DRONE WIND ANALYSIS

 **GO** Peak: 14.6/15 mph WSW

* Sunrise: 5:57 AM ☾ Sunset: 8:26 PM fly window clipped to daylight

 Best flight window: 6:00 AM – 9:00 PM (15 hrs)

 Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤turbulence	DIR
5 AM	6/5 mph		SSE
6 AM	5/4 mph		SE
7 AM	3/2 mph		SE
8 AM	1/2 mph		E
9 AM	3/5 mph		NNW
10 AM	6/9 mph		NNW
11 AM	9/13 mph		NW
12 PM	10/14 mph		NW
1 PM	11/15 mph		NW
2 PM	12/15 mph		WNW
3 PM	13/15 mph		WNW
4 PM	14/14 mph		W
5 PM	15/13 mph		W
6 PM	13/11 mph		WSW
7 PM	9/9 mph		WSW
8 PM	6/7 mph		SW

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS **G** ▶ FREE TO FLY
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→ TFR STATUS  **NO TFRs WITHIN 30 MI** [Check B4UFLY ↗](#)
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 90°F

Low 56°F

Wind max 13 mph

Gusts 15 mph

Precip 0.00"

Sunrise 5:58 AM

Sunset 8:25 PM

☾ Waning Gibbous 99%

GNSS: Avg PDOP **0.98** · Best 4-hr window 1:00 PM – 5:00 PM (PDOP 0.95, 33 sats avg)

GNSS / DOP - HOURLY			
	PDOP	SATS	RATING
6:00 AM	0.91	36 sv	IDEAL
7:00 AM	0.90	35 sv	IDEAL
8:00 AM	1.06	31 sv	GOOD
9:00 AM	1.03	30 sv	GOOD
10:00 AM	0.98	29 sv	IDEAL
11:00 AM	1.02	29 sv	GOOD
12:00 PM	1.09	27 sv	GOOD
1:00 PM	0.87	33 sv	IDEAL
2:00 PM	0.96	31 sv	IDEAL
3:00 PM	0.98	33 sv	IDEAL
4:00 PM	0.98	35 sv	IDEAL
5:00 PM	1.01	30 sv	GOOD
6:00 PM	0.97	31 sv	IDEAL
7:00 PM	0.93	30 sv	IDEAL



GO – GOOD CONDITIONS

Partly cloudy | Wind: 13.2 mph (gusts 15.2 mph) | Precip: 0" | Temp: 55.7°–89.5°F



DRONE WIND ANALYSIS



GO Peak: 13.2/15.2 mph W

* Sunrise: 5:58 AM ☾ Sunset: 8:25 PM fly window clipped to daylight

✓ Best flight window: 6:00 AM – 9:00 PM (15 hrs)

🔄 Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤turbulence	DIR
5 AM	6/5 mph		SSE
6 AM	4/3 mph		SSE
7 AM	2/1 mph		SE
8 AM	1/1 mph		NNW
9 AM	3/3 mph		NNW
10 AM	5/7 mph		NNW
11 AM	8/11 mph		NNW
12 PM	9/13 mph		NNW
1 PM	11/15 mph		NNW
2 PM	12/15 mph		NW
3 PM	12/15 mph		NW
4 PM	13/14 mph		WNW
5 PM	13/12 mph		W
6 PM	11/9 mph		W
7 PM	7/7 mph		W
8 PM	4/5 mph		WSW

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS G ▶ FREE TO FLY

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→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

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FIELD PLANNING REPORT — GNSS SURVEY OPERATIONS

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SITE INFORMATION

Location: 40.8709° N, 121.3448° W (Shasta County, CA)

Elevation: Baseline terrain; recommended elevation mask 13° (dwarf scrub, low obstruction)

Report Date: Wednesday, July 22, 2026

Survey Window: July 22–28, 2026 (7 days)

=====

IONOSPHERIC & SPACE WEATHER SUMMARY

=====

Peak Kp index: 5.0 (average 2.0) — ELEVATED GEOMAGNETIC ACTIVITY

Solar Flux F10.7: 138 sfu — nominal range

Vertical TEC: 18.4 TECU (below 30-day median by 2.0 TECU; data quality flag noted)

ASSESSMENT: Elevated ionospheric disturbance expected throughout the analysis window.

Anticipate increased ionospheric delays and elevated cycle-slip risk on all days.

Recommend extended session lengths (≥2 hr minimum) and conservative baseline acceptance criteria for precision static work. RTK/kinematic work will experience greater convergence delays. No day is exempt; all rankings reflect this baseline constraint.

=====

SATELLITE CONSTELLATION STATUS

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PRN 11 — UNHEALTHY (UNUSUFN) — no return date. Auto-excluded by receivers.

PRN 3 — Scheduled maintenance Thu 09 Jul 18:00 — outside survey window; informational.

PRN 26 — Scheduled maintenance Fri 17 Jul 00:35–12:35 UTC — outside survey window; informational.

PRN 7 — Scheduled maintenance Fri 24 Jul 01:15–13:15 UTC — no impact on day-time operations.

PDOP impact: Already factored into reported daily PDOP averages. No additional reduction.

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WEATHER FORECAST & CONDITIONS

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Wednesday 22 Jul: Overcast, H 84°F / L 65°F, wind max 15 mph, 0.00" precip

Thursday 23 Jul: Overcast, H 89°F / L 56°F, wind max 13 mph, 0.00" precip

Friday 24 Jul: CLEAR SKY, H 93°F / L 55°F, wind max 14 mph, 0.00" precip

Saturday 25 Jul: Overcast, H 90°F / L 54°F, wind max 16 mph, 0.00" precip [WEEKEND]

Sunday 26 Jul: CLEAR SKY, H 90°F / L 52°F, wind max 15 mph, 0.00" precip [WEEKEND]

Monday 27 Jul: Mainly clear, H 91°F / L 54°F, wind max 15 mph, 0.00" precip

Tuesday 28 Jul: Mainly clear, H 90°F / L 56°F, wind max 13 mph, 0.00" precip

No precipitation forecast for the entire window. Wind speeds benign for survey operations.

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WILDFIRE & ENVIRONMENTAL STATUS

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NIFC Active Fire Database: No active fires within detection range (Shasta County).

Visibility and smoke impact: CLEAR — all-clear status maintained.

=====

TODAY (WEDNESDAY, JULY 22) — DECISION BRIEF

=====

STATUS: CAUTION

RATIONALE:

- PDOP: Avg 0.98 (acceptable); best window 16:00–20:00 (PDOP 0.93, 33 sats)
- Weather: Overcast conditions; no precipitation
- Ionosphere: Elevated Kp = 5.0; expect significant delays and cycle slips
- Timing: Report distributed mid-morning; crews can mobilize for afternoon window only

RECOMMENDATION: Deploy for afternoon static session (16:00–20:00) only if high-precision baseline is not critical. Recommend full-length session (≥ 2 hr) with conservative QC thresholds. If work can wait, defer to Friday or Monday for better overall conditions.

=====

MONDAY CALL — CREW MOBILIZATION DECISION

=====

STATUS: GO — PROCEED WITH MOBILIZATION

RATIONALE:

- PDOP: Avg 0.98, best window 16:00–20:00 (PDOP 0.95, 32 sats) — solid geometry
- Weather: Mainly clear, ideal conditions
- Ionosphere: Kp still elevated but stable; TEC below median (favorable)
- Scheduling: Weekday slot; minimal operational friction

CREW CHIEF DECISION: Mobilize crew Sunday evening or early Monday. Plan primary static work 16:00–20:00 Monday (4-hour minimum session). Conditions are reliable; expect standard convergence timelines with elevated ionosphere constraint.

=====

7-DAY RANKING — BEST TO WORST FOR GNSS SURVEY CONDITIONS

=====

RANK 1 — FRIDAY, JULY 24

PDOP 0.97 avg, best window 12:00–16:00 (PDOP 0.91, 34 sats) | Clear sky | Ionosphere elevated
GRADE: EXCELLENT — best PDOP, best sky conditions, high satellite count. FIRST CHOICE.

RANK 2 — MONDAY, JULY 27

PDOP 0.98 avg, best window 16:00–20:00 (PDOP 0.95, 32 sats) | Mainly clear | Ionosphere stable
GRADE: EXCELLENT — weekday, reliable geometry, clear weather. SECOND CHOICE.

RANK 3 — SUNDAY, JULY 26 [WEEKEND]

PDOP 0.97 avg, best window 13:00–17:00 (PDOP 0.92, 35 sats) | Clear sky | Ionosphere stable
GRADE: VERY GOOD — strong PDOP, clear sky, high sat count. Weekend slot limits crew availability.

RANK 4 — SATURDAY, JULY 25 [WEEKEND]

PDOP 0.95 avg, best window 15:00–19:00 (PDOP 0.88, 36 sats) | Overcast | Ionosphere stable
GRADE: VERY GOOD — best average PDOP of week, high sat count. Weekend + overcast lower priority.

RANK 5 — TUESDAY, JULY 28

PDOP 0.98 avg, best window 13:00–17:00 (PDOP 0.95, 33 sats) | Mainly clear | Ionosphere stable
GRADE: GOOD — stable weekday, acceptable PDOP, clear weather.

RANK 6 — THURSDAY, JULY 23

PDOP 0.99 avg, best window 15:00–19:00 (PDOP 0.97, 32 sats) | Overcast | Ionosphere elevated Kp

GRADE: ACCEPTABLE — marginal PDOP, overcast conditions, elevated Kp.

RANK 7 — WEDNESDAY, JULY 22 (TODAY)

PDOP 0.98 avg, best window 16:00–20:00 (PDOP 0.93, 33 sats) | Overcast | Ionosphere PEAK Kp 5.0

GRADE: CAUTION — today only; peak geomagnetic disturbance, limited daylight window remaining.

TOP 5 RECOMMENDED WORK DAYS: Friday 24, Monday 27, Sunday 26, Saturday 25, Tuesday 28

=====

BEST STATIC WINDOW — WEEK OVERVIEW

=====

OPTIMAL STATIC SESSION: Friday, July 24, 12:00–16:00 (local time)

Duration: 4 hours continuous

Expected PDOP: 0.91–0.94

Satellite count: 34 (GPS + GLONASS + Galileo + BeiDou multi-constellation)

Sky conditions: Clear

Ionospheric constraint: Elevated Kp (5.0), but TEC favorable; expect standard delays

Expected convergence (static, baseline <5 km): 15–25 min (baseline-dependent)

Quality: EXCELLENT — lowest PDOP of week, clear sky, strong geometry

ALTERNATIVE WINDOW (if Friday unavailable): Monday, July 27, 16:00–20:00 (4 hours)

PDOP 0.95–0.98, 32 sats, mainly clear sky, weekday operations. High confidence.

=====

DRONE ASSESSMENT — DAY BY DAY

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Wednesday 22 Jul: CAUTION — Overcast, wind 15 mph max, brief daylight. Use AM only if critical.

Thursday 23 Jul: CAUTION — Overcast, wind 13 mph, marginal visibility for precision imaging.

Friday 24 Jul: FLYABLE — Clear sky, wind 14 mph, excellent visibility. RECOMMENDED.

Saturday 25 Jul: CAUTION — Overcast, wind 16 mph (gust potential), weekend ops friction.

Sunday 26 Jul: FLYABLE — Clear sky, wind 15 mph, strong visibility. RECOMMENDED (weekend).

Monday 27 Jul: FLYABLE — Mainly clear, wind 15 mph, excellent conditions. RECOMMENDED.

Tuesday 28 Jul: FLYABLE — Mainly clear, wind 13 mph, good visibility. RECOMMENDED.

PRIORITY DRONE DAYS: Friday 24, Sunday 26, Monday 27 (clear + low wind + daylight).

=====

FIELD WARNINGS — CRITICAL SAFETY & OPERATIONAL ISSUES

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1. IONOSPHERIC DISTURBANCE (Kp 5.0, elevated throughout)

COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Shasta County Assessor-Recorder

1450 Court St, Redding, CA 96001

ASSESSOR / TAX OFFICE

Shasta County Assessor

1450 Court St, Redding, CA 96001

This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning

purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
 NGS ↗	P348	HATCHETMTNCN2005	25.4 mi
 NGS ↗	P347	ADINCTYARDCN2007	29.9 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet ↗	LT0642	"5582=BALD MTN"	1700.900m / 5580.4ft NAVD88	Order	2.7 mi
 Datasheet ↗	LT0643	"BALD MOUNTAIN"	1702.300m / 5585.0ft NAVD88	Order	2.7 mi
 Datasheet ↗	LT0637	"LADDER RM"	1751.000m / 5744.8ft NAVD88	Order	4.6 mi
 Datasheet ↗	LT0636	"LADDER"	1751.300m / 5745.7ft NAVD88	Order	4.6 mi
 Datasheet ↗	LT0638	"RIM"	1562.100m / 5125.0ft NAVD88	Order	5.6 mi
 Datasheet ↗	LT0641	"HAT CREEK RIM LOT"	1562.000m / 5124.7ft NAVD88	Order	5.7 mi
 Datasheet ↗	LT0227	"3897.9"	1189.340m / 3902.0ft NAVD88	Order	6.9 mi
 Datasheet ↗	LT0226	"M"	1189.580m / 3902.8ft NAVD88	Order 1	6.9 mi

FEMA FLOOD ZONE

D Panel: N/A
Effective: N/A
[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

Download .txt File

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector
Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
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LT0642	5582=BALD MTN	40.900699	-121.378314	5580.4	
LT0643	BALD MOUNTAIN	40.900708	-121.378332	5585.0	
LT0637	LADDER RM	40.810583	-121.306819	5744.8	
LT0636	LADDER	40.810546	-121.306902	5745.7	
LT0638	RIM	40.808727	-121.414863	5125.0	
LT0641	HAT CREEK RIM LOT	40.808552	-121.414828	5124.7	
LT0227	3897.9	40.927222	-121.454167	3902.0	
LT0226	M	40.927222	-121.454722	3902.8	1

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Shasta Lake	57.0 mi	1,017.81 ft (-49 ft)	1,067 ft	3,236K af	71%
Black Butte Lake	89.8 mi	—	228 ft	68K af	50%
Lake Oroville	92.4 mi	850.13 ft (-50 ft)	900 ft	2,710K af	77%
Folsom Lake	149.7 mi	441.79 ft (-24 ft)	466 ft	724K af	74%

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Whiskeytown Lake	65 mi	1,209 ft*	1,209 ft	238K af	99%
Trinity Lake	74 mi	2,370 ft*	2,370 ft	1,926K af	79%

ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Black Butte-Pool	100 mi	455.2 ft	458.36 ft	—	99%
Englebright Lake-Pool	113 mi	524.5 ft	522.35 ft	—	100%
Boca	134 mi	5,600.9 ft	5,579.88 ft	—	100%

SNOTEL STATIONS (NRCS) – within 50 miles

STATION	DIST	ELEV	SWE	DEPTH	TEMP	ACCESS
Blacks Mountain [Snow Course]	10.4 mi	7,060 ft	—	—	—	● Clear
Blacks Mountain [Snow Course]	10.9 mi	6,770 ft	—	—	—	● Clear
Mcelroy Pass [Snow Course]	14.1 mi	5,100 ft	—	—	—	● Clear
Burney Springs [Snow Course]	15.6 mi	4,700 ft	—	—	—	● Clear
Thousand Lakes [Snow Course]	16.5 mi	6,480 ft	—	—	—	● Clear

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
035N007E15P001M	8.6 mi	300 ft	—	—	N9999OTHER
034N004E15M001M	10.1 mi	290 ft	206.17 ft BGS	2020-10-07	N100PCFNWV
037N005E19P002M	11.6 mi	225 ft	—	—	N100PCFNWB
035N004E06J001M	11.7 mi	244 ft	—	—	N100PCFNWV
037N004E11A001M	14.7 mi	185 ft	26.20 ft BGS	1983-04-14	N100PCFNWB
035N003E03L001M	15 mi	226 ft	—	—	N100PCFNWV
037N007E02D001M	16.9 mi	60 ft	—	—	N100PCFNWB

○ AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

This site overlies a mixed aquifer system in the Pit River–Hat Creek basin of northeastern California, composed of Quaternary alluvium and volcanic deposits (basalt, andesite) that filled paleovalleys and low areas during repeated glacial cycles. These materials were deposited by rivers, glacial melt, and volcanic activity over the past ~2 million years, creating layers of sand, gravel, and fractured rock with moderate to good water-yielding capacity. Water availability is moderate but variable with depth; shallow zones (60–226 ft) are reliably productive, while deeper zones show significant water-level decline, indicating drawdown stress or reduced recharge at depth.

FORMATION

Name	Quaternary alluvium and undifferentiated volcanic deposits (Pit River–Hat Creek basin aquifer complex)
Age / Rock	Quaternary (~2 million to 10,000 years ago); sand, gravel, basalt, and andesite with interbedded clay
Extent	800–1200 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	60–300 ft (6–30 floors)
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Typical Depth	100–200 ft (10–20 floors)
Recharge Source	Precipitation on surrounding uplands and basin margins, lateral inflow from volcanic highlands, seasonal snowmelt
Transit Time	20–50 years to 100 ft; 50–150 years to 200+ ft depth
WATER QUALITY	
pH	est. 6.8–7.6 (est. USGS/DWR)
Hardness	est. 150–350 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 250–550 mg/L (est. USGS/DWR)
Notes	No sample data available for this site. Typical for formation: moderately hard, neutral pH, low to moderate salinity. Volcanic aquifers regionally free of industrial contaminants but check for agricultural nitrate in shallow zones.
TREND	
Trend	Significant water-level decline over 37 years: from 26.20 ft BGS (April 1983) to 206.17 ft BGS (October 2020), a drawdown of ~180 ft. This indicates sustained groundwater extraction exceeding recharge.
Source	USGS NWIS well logs and DWR historical records
Seasonal Swing	est. 8–20 ft seasonal range (est. USGS/DWR); higher in spring (snowmelt), lower in late summer–fall

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ **LOW SUBSIDENCE RISK — No known critically overdrafted basin**
Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🕒 AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU	
The site is located in northeastern California near the Modoc Plateau, an area characterized by consolidated volcanic and metamorphic bedrock with minimal clay-rich aquitard sequences. The well-drained soil classification and absence from any critically overdrafted groundwater basin indicate stable aquifer conditions without significant groundwater depletion history. Surveyors working in this area can rely on published benchmarks with confidence, as the geological setting does not support the clay compaction and aquifer drawdown mechanisms that drive subsidence in California's Central Valley and other alluvial basins.	
SUBSIDENCE ASSESSMENT	
Susceptibility	Low
Primary Mechanism	None - consolidated terrain
Annual Rate	<0.5 mm/yr (est.)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in SGMA basin
BENCHMARK IMPLICATIONS	
Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence
GNSS Recommendation	Standard benchmark verification adequate
SURVEY ACTION	
Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)		WIND / SNOW / ICE / FROST (ASCE 7-22)	
Site Class	D (default)	Basic Wind Speed	95 mph
Risk Category	II	Ground Snow Load	80.3 psf
		Winter Wind (W2)	0.35
		Ice Thickness	0.11 in
		Frost Depth	30 in minimum
		Brown's Formula · C=1.2 (Sandy loam, Group B) · AFI=356°F-days · SF=1.25	
		Verify with local building official before construction	
		Source: ASCE 7-22 Hazard Tool · gis.asce.org	

CLIMATE ZONE (IECC 2021): Zone 3B · Hot-Dry · Hot-Dry - warm/hot summers, cool winters, dry (Central CA, NV, AZ) County: Shasta

🕒 AI CLIMATE SUMMARY — RESIDENTIAL DESIGN	
This hot-dry climate demands a well-insulated, air-tight envelope with minimal air leakage, since cooling loads dominate and humidity is rarely a concern. Prioritize glazing on the north and east facades while minimizing west and south-facing glass, or use deep overhangs and external shading on southern exposures to block summer heat; the high wind speed of 95 mph also requires careful attention to window placement and structural bracing, and the substantial 80.3 psf snow load is unusual for this region and suggests either high elevation or a data anomaly that warrants verification with the local building official. The most effective passive strategy for this zone is to use thermal mass (concrete slabs, masonry walls) on south-facing or internally shaded areas to absorb daytime heat and passively release it at night when outdoor temperatures drop significantly.	
🕒 AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU	
This site in central California is anticipated to have well-drained soils suitable for conventional spread footing foundations, with moderate snow and wind loads that may warrant standard structural detailing. Seismic parameters are currently null and may warrant clarification with the local building official; consult a licensed engineer to confirm applicable design standards and soil-structure	

interaction for this Site Class D location.

FOUNDATION

Type	Conventional spread footing anticipated, subject to geotechnical investigation
Basis	Well-drained Brownlee-Bobbitt soils (Hydro Group B) and 30-inch frost depth suggest conventional foundation design may be feasible; however, consult engineer to confirm bearing capacity and settlement characteristics

STRUCTURAL IMPLICATIONS

Seismic	Seismic design coefficients currently null — ordinary detailing may be adequate pending clarification of SDC; recommend confirming with local authority having jurisdiction
Wind Exposure	Exposure C (open terrain) suggested for ASCE 7-22 — verify final site conditions and nearby obstructions
Wind	95 mph 3-second gust wind load anticipated — standard wind load design likely adequate, no hurricane exposure
Snow	80.3 psf ground snow load anticipated — standard roof design and slope consideration likely adequate; winter wind factor W2=0.35 may warrant consultation on drift potential

INVESTIGATIONS & INSPECTIONS

Geotech Required	Yes — geotechnical investigation recommended to confirm Site Class D designation, bearing capacity, settlement estimates, and frost depth verification with local building official
Liquefaction Screen	Low risk anticipated — site not in known subsidence zone and Hydro Group B suggests lower groundwater; however, subsurface exploration may warrant screening confirmation
Special Inspections	Not anticipated for current design parameters; may be required if seismic classification is clarified — consult engineer and local authority
Code Edition	ASCE 7-22 / IBC 2021 (planning reference only)
Site Class Note	Site Class D assumed for planning — geotechnical investigation and local building official consultation recommended to confirm site classification and refine foundation design

SOILS — USDA SOIL SURVEY

MAP UNIT:	Brownlee-Bobbitt families association, 0 to 35 percent slopes.	SERIES:	Brownlee
DRAINAGE CLASS:	Well drained	HYDRO GROUP:	B
SOIL ORDER:	Mollisols	SUBORDER:	Xerolls

AI SOIL INTELLIGENCE — CLAUDE HAIKU

Brownlee soils are well-drained Mollisols (Xerolls) formed in mixed alluvial and colluvial material on moderate slopes in the semi-arid interior Pacific Northwest. These soils feature a dark mollic epipedon over a clay loam or gravelly clay loam subsoil, reflecting their formation under sagebrush steppe vegetation in a dry climate. Site personnel should anticipate moderate slopes with variable gravel content, generally good bearing capacity, and stable excavation conditions, though seasonal moisture fluctuations and clay substratum properties warrant attention during design and construction.

Mixed alluvial and colluvial parent material on hillslopes and footslopes; sagebrush steppe vegetation.

DRAINAGE

Class	Well drained
Seasonal Saturation	None within 60 inches
Flooding Risk	Minimal on 0–3% slopes; low to moderate on steeper terrain depending on position
Ponding Risk	None

ENGINEERING

Bearing Capacity	3,500–4,500 psf (shallow foundations)
Rating	Good
Shrink-Swell	Moderate — clay loam to gravelly clay loam substratum
Frost Action	Low to moderate
Cut Suitability	Good (gravel content aids drainage and stability)
Fill Suitability	Fair (clay substratum — moisture conditioning required)

SEPTIC

Suitability	Conditional
Notes	Perc test required — clay loam subsoil and variable gravel content may restrict percolation; slope and seasonal water table must be verified

HAZARDS

Excavation	Moderate — 15–35% gravel and cobble content; rippable but may require equipment selection
Expansive Clay	None anticipated
Erosion	Low on 0–15% slopes; moderate to high on steeper slopes and in disturbed areas; gully initiation risk in concentrated runoff zones
Other	None

Data sources: California DWR / CDEC · cdec.water.ca.gov · USBR RISE · data.usbr.gov · USACE CWMS · water.usace.army.mil · NRCS SNOTEL/AWDB · wcc.sc.egov.usda.gov · USGS NWIS Groundwater · waterservices.usgs.gov · USDA Web Soil Survey · sdataaccess.nrcs.usda.gov · CA DWR SGMA Bulletin 118 · gis.water.ca.gov · ASCE 7-22 Hazard Tool · gis.asce.org · USGS Design Maps · earthquake.usgs.gov · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision
Water levels current as of report run: Jul 22, 2026, 2:55 AM UTC
* Live pool elevation not reported by this reservoir — full pool design elevation shown

 OIL & GAS WELLS California — within 10mi radius

No oil or gas wells found within 10 miles per California public data.

AI Assessment — Well Proximity

No wells in area.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

127
total in search area

1
within 5 mi

48
within 25 mi

Site Name	Status	Commodities	Distance
Cinder Pit	Producer	Sand/Gravel	4.9 mi
Cinder Pits	Unknown	Stone/Crushed	6 mi
Cinder Pits	Occurrence	STN	6.4 mi
Six Mile Cinder Pit and Mill	Past Producer	Stone/Crushed	7 mi
Fall River Quarry B	Producer	PUM	7.3 mi
Pit River Pit	Producer	Sand/Gravel	8.5 mi
Wilcox Pit & Mill	Producer	PUM	11.1 mi
Grefco	Past Producer	DIT	12 mi
Hat Creek Pit	Producer	Sand/Gravel	12.7 mi
Sand Pits	Occurrence	Sand/Gravel	13.1 mi
Brush Mountain Quarry	Producer	PUM	13.2 mi
Gravel Pit	Unknown	Sand/Gravel	14.3 mi
Sand Pits	Occurrence	Sand/Gravel	14.4 mi
Venture II Mine	Producer	DIT	14.5 mi
Unnamed Property	Occurrence	DIT	14.7 mi
Quarry	Occurrence	Stone/Crushed	14.7 mi
Quarry	Unknown	Stone/Crushed	14.9 mi
Burney Pit	Producer	DIT	15.5 mi
Cinder Pit	Unknown	Stone/Crushed	16.4 mi
Cinder Pits	Unknown	Stone/Crushed	16.5 mi
Cinder Pit	Occurrence	STN	16.6 mi
Gravel Pit	Unknown	Sand/Gravel	16.6 mi
Cinder Pits	Occurrence	STN	16.8 mi
Cinder Pit	Unknown	Stone/Crushed	17.1 mi
Cinder Pit	Occurrence	STN	17.2 mi

Showing 25 of 127 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

The Cinder Pit operation (4.9 miles away) is the only active producer within 5 miles of your site. Expect heavy truck traffic on access roads during operating hours and potential dust generation typical of sand and gravel operations. No historical deposits warrant Phase 1 assessment consideration based on proximity.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

1
Endangered

1
Threatened

2
Total w/ Critical Habitat

1 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
Greene's tuctoria	Tuctoria greenei	Endangered

COMMON NAME	SCIENTIFIC NAME	STATUS
Slender Orcutt grass	<i>Orcuttia tenuis</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Implications Summary The presence of two listed species within 25 miles of the subject property (40.8709, -121.3448) triggers ESA Section 7 consultation requirements if federal agency involvement or federal permits/funding are anticipated for project activities. Greene's tuctoria (endangered) represents the most planning-relevant species given its higher protection status and potential habitat sensitivity in this northeastern California region. A Phase I biological resource assessment is recommended as the immediate next step to determine site-specific habitat suitability, baseline species presence, and project impact potential to inform agency consultation strategy and feasibility analysis. Early coordination with the U.S. Fish and Wildlife Service during project scoping will clarify consultation pathways and minimize permitting delays. **For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.**

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

66	242	2026-07-07
Unique Species	Total Observations	Most Recent
COMMON NAME	SCIENTIFIC NAME	LAST OBS
Antelope Bitterbrush	<i>Purshia tridentata</i>	2026-05-12
Bach's calicoflower	<i>Downingia bacigalupii</i>	2025-06-19
Bear Valley buckwheat	<i>Eriogonum ursinum</i>	2025-06-19
bighead clover	<i>Trifolium macrocephalum</i>	2024-05-07
bractless hedge-hyssop	<i>Gratiola ebracteata</i>	2025-06-19
Bulbous woodland star	<i>Lithophragma glabrum</i>	2024-05-21
California damasonium	<i>Damasonium californicum</i>	2026-05-12
Chinese caps	<i>Euphorbia crenulata</i>	2024-05-30
Cobwebby Thistle	<i>Cirsium occidentale</i>	2024-05-30
common hareleaf	<i>Lagophylla ramosissima</i>	2023-07-09
Common Sunflower	<i>Helianthus annuus</i>	2025-07-02
Common Yellow Chaenactis	<i>Chaenactis glabriuscula</i>	2026-05-12
Curlleaf Mountain Mahogany	<i>Cercocarpus ledifolius</i>	2026-05-12
cut-leaved paintbrush	<i>Castilleja lacera</i>	2025-06-18
Doublehorn Calicoflower	<i>Downingia bicornuta</i>	2026-05-12
Douglas' stitchwort	<i>Sabulina douglasii</i>	2026-05-12
dwarf hesperochiron	<i>Hesperochiron pumilus</i>	2024-05-21
field owl's-clover	<i>Castilleja campestris</i>	2026-05-12
flatbud prickly poppy	<i>Argemone munita</i>	2025-06-19
folded calicoflower	<i>Downingia ornatissima</i>	2026-05-12
fragrant sumac	<i>Rhus aromatica</i>	2026-05-12
grand collomia	<i>Collomia grandiflora</i>	2025-06-05
gray pine	<i>Pinus sabiniana</i>	2026-05-12
greenleaf manzanita	<i>Arctostaphylos patula</i>	2025-04-20
heart-leaf milkweed	<i>Asclepias cordifolia</i>	2026-05-12
Layne's monkeyflower	<i>Diplacus layneae</i>	2025-06-19
Linearleaf Phacelia	<i>Phacelia linearis</i>	2026-05-23
Long-stalked Clover	<i>Trifolium longipes</i>	2024-05-21
Low Pussytoes	<i>Antennaria dimorpha</i>	2024-05-29
Mahala mat	<i>Ceanothus prostratus</i>	2024-05-29
-	<i>Navarretia leucocephala leucocephala</i>	2026-05-12
Nevada lewisia	<i>Lewisia nevadensis</i>	2024-05-30
oblong bluecurls	<i>Trichostema oblongum</i>	2025-06-18
one-spike oat grass	<i>Danthonia unispicata</i>	2025-06-19
Oregon oak	<i>Quercus garryana</i>	2026-07-07
Oregon woollyheads	<i>Psilocarphus oregonus</i>	2025-06-04
pineapple-weed	<i>Matricaria discoidea</i>	2026-05-12
plains evening primrose	<i>Camissonia contorta</i>	2026-05-12
prairie smoke	<i>Geum triflorum</i>	2024-05-29
Purslane Speedwell	<i>Veronica peregrina</i>	2024-05-21
Red larkspur	<i>Delphinium nudicaule</i>	2026-05-12

COMMON NAME	SCIENTIFIC NAME	LAST OBS
rock buckwheat	<i>Eriogonum sphaerocephalum</i>	2023-07-08
rough eyelashweed	<i>Blepharipappus scaber</i>	2024-05-30
Round-Tooth Ookow	<i>Dichelostemma multiflorum</i>	2026-05-12
Rydberg's Beardtongue	<i>Penstemon rydbergii</i>	2024-05-29
Sagebrush Mariposa Lily	<i>Calochortus macrocarpus</i>	2025-07-03
Siskiyou bluecurls	<i>Trichostema simulatum</i>	2023-07-09
small tarweed	<i>Madia exigua</i>	2026-05-12
Spineless Horsebrush	<i>Tetradymia canescens</i>	2023-07-09
spotted fritillary	<i>Fritillaria atropurpurea</i>	2024-05-30
spreading dogbane	<i>Apocynum androsaemifolium</i>	2025-06-19
springbank clover	<i>Trifolium wormskoldii</i>	2025-06-19
Tehama navarretia	<i>Navarretia heterandra</i>	2026-05-12
thread-leaf fleabane	<i>Erigeron filifolius</i>	2025-07-03
toothed calicoflower	<i>Downingia cuspidata</i>	2026-05-12
tracy's eriastrum	<i>Eriastrum tracyi</i>	2025-06-19
tricolor monkeyflower	<i>Diplacus tricolor</i>	2026-05-12
wavyleaf paintbrush	<i>Castilleja applegatei</i>	2024-05-29
western blue flag	<i>Iris missouriensis</i>	2025-06-19
western columbine	<i>Aquilegia formosa</i>	2025-06-19
western juniper	<i>Juniperus occidentalis</i>	2026-05-12
Western Redbud	<i>Cercis occidentalis</i>	2023-06-21
whitehead navarretia	<i>Navarretia leucocephala</i>	2026-05-12
Willow Moss	<i>Fontinalis antipyretica</i>	2025-06-19
wirestem	<i>Rigiopappus leptocladus</i>	2026-05-12
woolly fishhooks	<i>Ancistrocarphus filagineus</i>	2026-05-12

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment: 40.8709, -121.3448 This site supports a **semi-arid shrubland and grassland mosaic**, dominated by drought-adapted species like Antelope Bitterbrush, Curlleaf Mountain Mahogany, and various native wildflowers characteristic of the interior Pacific Northwest. The ecology reflects intermittent moisture availability, with seasonal wetland indicators (California damasonium, bractless hedge-hyssop, dwarf hesperochiron) suggesting vernal pools or seepage areas that support ephemeral specialists alongside upland xeric species. **Soil moisture conditions are highly variable across the site:** well-drained, rocky uplands support shrubs and deep-rooted perennials, while localized low areas retain seasonal moisture critical for calicoflower and stitchwort populations. For grading and drainage planning, avoid disrupting these moisture-dependent pockets, as they support rare annuals and support groundwater recharge; implement drainage infrastructure that preserves natural water infiltration patterns in low-lying areas. **Consult a licensed botanist for site-specific vegetation assessment.**

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

10	3	6	6
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants
VENOMOUS SNAKES			
SPECIES		OBSERVATIONS	
Crotalus oreganus oreganus		10 obs	
Crotalus oreganus		10 obs	
VENOMOUS SPIDERS			
SPECIES		OBSERVATIONS	
Latrodectus hesperus		3 obs	
STINGING INSECTS			
SPECIES		OBSERVATIONS	
Dasymutilla californica		6 obs	
Dasymutilla aureola		6 obs	
TOXIC PLANTS			
SPECIES		OBSERVATIONS	
Toxicodendron diversilobum		5 obs	
Conium maculatum		1 obs	

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Brief – Shasta County Area **Highest-priority hazards:** Northern Pacific rattlesnakes are your most dangerous concern in this region—they're common, venomous, and often encountered during ground-level survey work; poison oak is nearly everywhere and causes severe skin reactions on contact. **Avoidance:** Wear snake boots or gaiters, watch where you place hands and feet, stay on cleared paths when possible, and learn to identify poison oak's three-leaflet pattern to avoid brushing against it; tuck pants into boots and wear long sleeves despite heat. **Seasonal note:** Rattlesnakes are most active April through October when temperatures warm, so heighten vigilance during spring and summer fieldwork; poison oak remains a threat year-round but causes worse reactions in warmer months when oils are more potent. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FAA AIRWAYS — ATS Routes crossing site area

17	7	7	3
Total Airways	Jet Routes	Victor Airways	Other Routes

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	342.8°	161.8°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J189	Jet Route	High	18,000 ft	—	335.5°	155.0°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J65	Jet Route	High	18,000 ft	—	325.5°	144.0°
J67	Jet Route	High	18,000 ft	—	347.7°	166.1°
J92	Jet Route	High	18,000 ft	—	131.4°	313.4°
Q11	Q-Route	High	—	—	155.1°	336.1°
Q7	Q-Route	High	—	—	155.3°	336.8°
Q9	Q-Route	High	—	—	155.9°	337.1°
V195	Victor Airway	Low	3,000 ft	—	320.2°	140.1°
V199	Victor Airway	Low	3,000 ft	—	19.7°	199.8°
V23	Victor Airway	Low	3,000 ft	—	320.2°	140.1°
V25	Victor Airway	Low	3,000 ft	—	4.6°	184.7°
V332	Victor Airway	Low	6,000 ft	—	301.9°	121.7°
V452	Victor Airway	Low	9,000 ft	14,000 ft	131.6°	311.9°
V87	Victor Airway	Low	3,000 ft	—	341.2°	161.2°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

4
Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
BNSF	S	1	No	No	—
BNSF	Other	1	No	No	—
BNSF	Mainline	1	No	No	GATEWAY
Unknown	A	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200	88	69	8	10	1
Total Observations	Total Species	Birds	Mammals	Reptiles	Amphibians

BIRDS (69 species)

COMMON NAME	Scientific Name	OBS
Red-tailed Hawk	Buteo jamaicensis	10
American Robin	Turdus migratorius	7
California Scrub-Jay	Aphelocoma californica	7
American Kestrel	Falco sparverius	7
Song Sparrow	Melospiza melodia	5
American Goldfinch	Spinus tristis	5
Great Blue Heron	Ardea herodias	5
Common Goldeneye	Bucephala clangula	5
Bufflehead	Bucephala albeola	5

COMMON NAME	Scientific Name	OBS
Western Meadowlark	<i>Sturnella neglecta</i>	5
Mallard	<i>Anas platyrhynchos</i>	4
Ferruginous Hawk	<i>Buteo regalis</i>	4
Northern Harrier	<i>Circus hudsonius</i>	4
Rough-legged Hawk	<i>Buteo lagopus</i>	4
Turkey Vulture	<i>Cathartes aura</i>	3

+ 54 additional species

MAMMALS (8 species)

COMMON NAME	Scientific Name	OBS
Common Golden-mantled Ground Squirrel	<i>Callospermophilus lateralis</i>	4
Mule Deer	<i>Odocoileus hemionus</i>	3
Townsend's Big-eared Bat	<i>Corynorhinus townsendii</i>	2
Black-tailed Jackrabbit	<i>Lepus californicus</i>	2
American Pronghorn	<i>Antilocapra americana americana</i>	1
Pronghorn	<i>Antilocapra americana</i>	1
North American River Otter	<i>Lontra canadensis</i>	1
Western Gray Squirrel	<i>Sciurus griseus</i>	1

REPTILES (10 species)

COMMON NAME	Scientific Name	OBS
Western Fence Lizard	<i>Sceloporus occidentalis</i>	11
Gopher Snake	<i>Pituophis catenifer</i>	4
Sierra Garter Snake	<i>Thamnophis couchii</i>	3
Western Skink	<i>Plestiodon skiltonianus</i>	2
Northern Pacific Rattlesnake	<i>Crotalus oreganus oreganus</i>	2
Common Sagebrush Lizard	<i>Sceloporus graciosus</i>	1
Common Garter Snake	<i>Thamnophis sirtalis</i>	1
Western Yellow-bellied Racer	<i>Coluber constrictor mormon</i>	1
Western Terrestrial Garter Snake	<i>Thamnophis elegans</i>	1
Northwestern Pond Turtle	<i>Actinemys marmorata</i>	1

AMPHIBIANS (1 species)

COMMON NAME	Scientific Name	OBS
Pacific chorus frog	<i>Pseudacris regilla</i>	2

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment This site in northeastern California (near Lassen Peak region) supports a diverse wildlife community characteristic of mixed shrub-steppe and oak woodland transitional habitat, with notable riparian connectivity evidenced by waterfowl (Common Goldeneye, Bufflehead, Great Blue Heron) and the presence of North American River Otter and Northwestern Pond Turtle, indicating perennial or seasonal water features within the survey area. The mammalian assemblage—including American Pronghorn, Mule Deer, and Black-tailed Jackrabbit—reflects open grassland and sagebrush components, while the robust reptile diversity (10 species) and presence of Townsend's Big-eared Bat suggest minimally to moderately disturbed conditions with intact structural habitat complexity. Of particular regulatory concern, American Pronghorn and Northwestern Pond Turtle warrant survey attention prior to site development, as both are sensitive to habitat fragmentation and may indicate the presence of critical movement corridors or wetland/aquatic features subject to California environmental protections; the diverse garter snake community and Pacific chorus frog presence further indicate that seasonal wetlands or water retention areas support breeding amphibian populations. Overall, this site represents functional, relatively intact shrub-steppe and riparian-adjacent habitat supporting species composition consistent with low-to-moderate anthropogenic disturbance. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

✳️ PASSIVE + ACTIVE SOLAR PLANNING — Lat 40.87°

1 · AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.5 hrs	Overall solar resource
Summer average	8.9 hrs	June–August
Winter average	3.8 hrs	December–February
Worst month (Dec)	3.1 hrs	Size battery storage to this

2 · IDEAL SOLAR ARRAY PITCH

Optimal tilt = 40.9° (latitude) → 10/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	96.6%	good
5/12	22.6°	97.3%	good
6/12	26.6°	97.9%	good
7/12	30.3°	98.4%	→ excellent

8/12	33.7°	98.9%	↔ excellent
9/12	36.9°	99.4%	↔ excellent
10/12	39.8°	99.8%	↔ annual optimal
12/12	45°	99.4%	↔ excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +712 kWh/yr (~\$107 value/yr). Over 25yr panel life: ~\$2669 total gain.

3 • SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	47	30	Minimize south glass
South glass (SHGC 0.6)	71	45	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	25	103	△ Minimize — afternoon load
Roof — dark (absorb 0.95)	228	301	Highest load surface
Roof — light (absorb 0.35)	84	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	60	79	Best roof option for heat
South wall — dark	88	68	
South wall — light	29	23	Light color = major savings
Hardscape — asphalt	181	301	△ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr-ft² additional load per 100sf of adjacent hardscape.

Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 • ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 • THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3B Hot-Dry: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 • TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.2	1	1.2	OK
20ft	All year	2.2	2	2.5	OK
30ft	All year	3.2	3	4	Zone 2 check
40ft	All year	4.2	3.9	5.8	△ verify
50ft	All year	5	4.8	9.1	△ verify
60ft	All year	5.8	5.7	9.1	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.

Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.

△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 • ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45

Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45

Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor

†Balanced = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)

WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.

△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

☑ AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar & Energy Plan Your location gets about 6.5 hours of useful sunlight per day on average—think of it like having a solar panel work at full power for that many hours, which is excellent for generating electricity year-round, especially in summer when you hit nearly 9 hours. Your roof should tilt at roughly a 10-over-12 pitch (that's moderately steep) to catch the sun's rays most efficiently. Here's what surprises most people: a dark roof acts like a heat-absorbing blanket in your hot summers, pulling in over 300 BTU of heat per hour per square foot, but switching to a light-colored roof cuts that nearly in half—so that single choice can dramatically reduce your cooling costs, which really matters in a climate like yours. Your west-facing windows are a particular problem in afternoon heat; they'll gain 103 BTU per hour per square foot in summer, making that side of your home an oven, so you'll want good shading or reflective glass there. Here's where trees become your secret weapon: a 40-foot deciduous tree on your south side will shade your home for about 4 hours on summer days while still letting the sun through in winter when you need

warmth—though if you're in a wildfire-prone area, you'll want to balance that benefit against defensible space requirements. Thermal mass is simply the ability of heavy materials like concrete or thick walls to absorb heat during the day and release it slowly at night, keeping temperatures steadier. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.



SITE INTELLIGENCE SUMMARY

AI synthesis — Claude Sonnet

⚙️ Technical

👤 Client

Technical Brief — for architects, engineers, and surveyors

SITE INTELLIGENCE BRIEFING **40.87092°N, 121.34479°W | Shasta County, California**

Structural Design Drivers

Wind and snow load data are not available for this location; seismic parameters are similarly absent from the current dataset. This absence does not eliminate seismic risk in Shasta County—California building code will govern—but it does mean the design team cannot yet rank these loads against one another or establish which dominates the lateral system. Request USGS seismic hazard maps and local ground motion parameters (PGA, spectral accelerations) immediately. The climate classification IECC Zone 3B (Hot-Dry) with an Annual Freezing Index of 356°F-days indicates moderate but measurable winter freeze-thaw cycling, which will govern interior thermal design and mechanical planning but does not typically drive structural member sizing in this zone. Until seismic and wind data are populated, structural system selection should remain provisional.

Geotechnical Foundation Design

Soils across the site belong to the Brownlee-Bobbitt families association with slopes ranging 0 to 35 percent and are classified Hydrologic Group B (moderate infiltration, moderate runoff). These soils are well-drained, which is favorable for foundation stability and reduces standing water risk, but soil texture has not been specified and must be confirmed in phase II exploration. More critically, frost depth is established at 30 inches (derived from Brown's Formula using the measured AFI of 356°F-days). All footings in areas subject to frost action must bear below 30 inches or employ frost-protected shallow foundation (FPSF) design per current building code. Ground exploration should verify this frost depth estimate across the development footprint and confirm actual soil density, bearing capacity, and potential for differential settlement on sloped terrain.

Hazard Assessment

No Quaternary faults exist within the search radius, eliminating surface fault rupture as a primary design hazard. Subsidence basin data is absent, indicating either no known subsidence risk or insufficient data; this should be verified with the U.S. Geological Survey and the California Department of Conservation. Wind and snow data gaps remain unaddressed. Wildfire hazard data is not provided; given Shasta County's fire-prone character, request defensible space and wildland-urban interface (WUI) criteria and consider ember-resistant design features if the site falls within State Responsibility Area (SRA) lands.

Site Context and Mining Activity

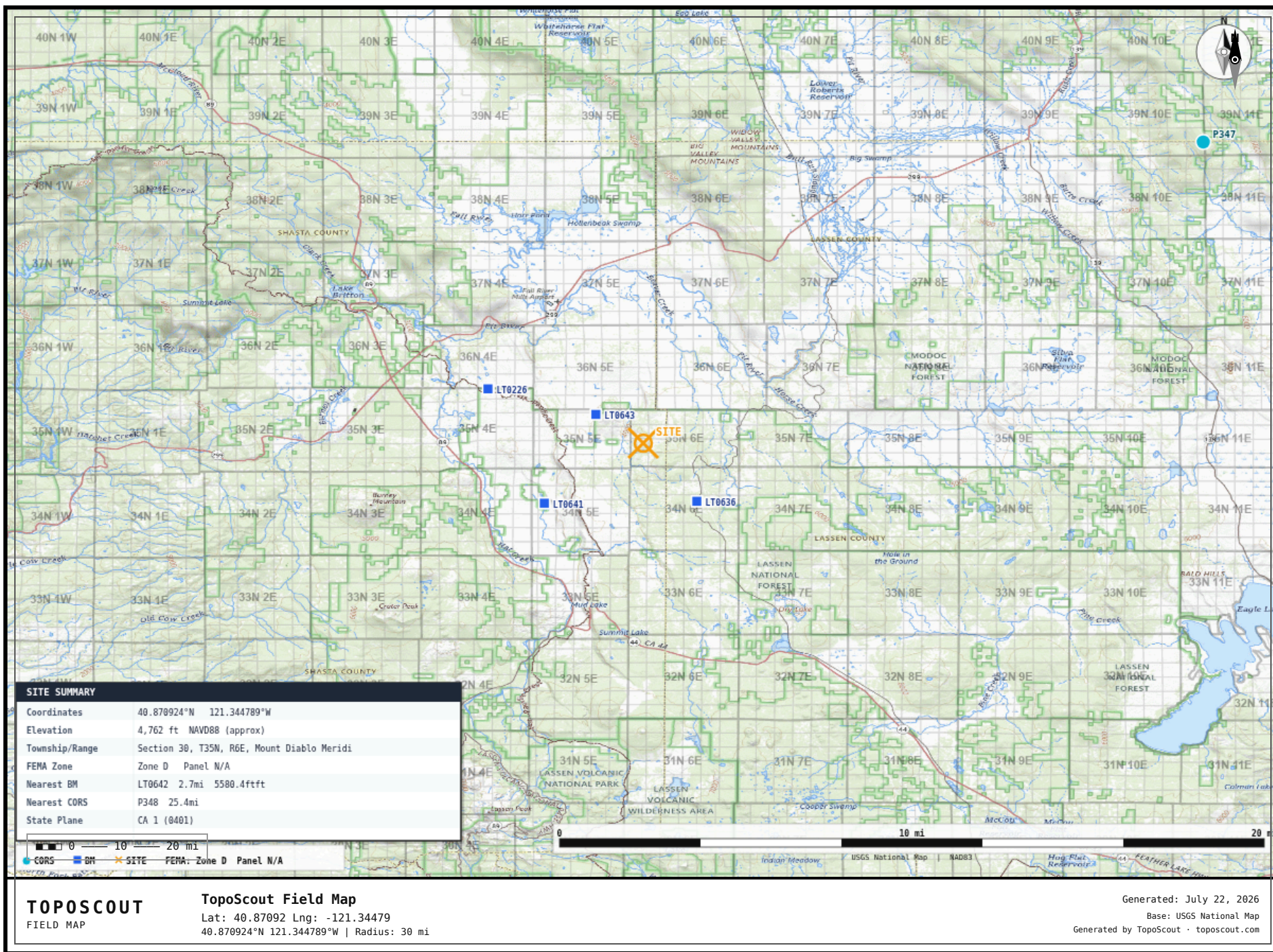
The site lies 4.9 miles from Cinder Pit, an active mineral aggregate producer (SDG commodity), and falls within a mineral-rich region hosting 127 MRDS-documented deposits within approximately 35 miles. Proximity to an active mine warrants confirmation of current and projected extraction boundaries, blast vibration protocols, noise and dust attenuation, and haul route impacts on site access and traffic. No oil or gas wells are recorded within the search radius per CalGEM data, eliminating that subsurface hazard. The Hot-Dry climate zone will drive aggressive mechanical cooling demand and daylighting/shading strategy but does not constrain foundation or structural layout.

Bottom line: Obtain seismic design parameters (PGA, site class, spectral response), confirm frost depth and soil bearing capacity through Phase II subsurface exploration, and coordinate with Cinder Pit operators on mine activity status and vibration/dust protocols before finalizing site plan and foundation strategy.

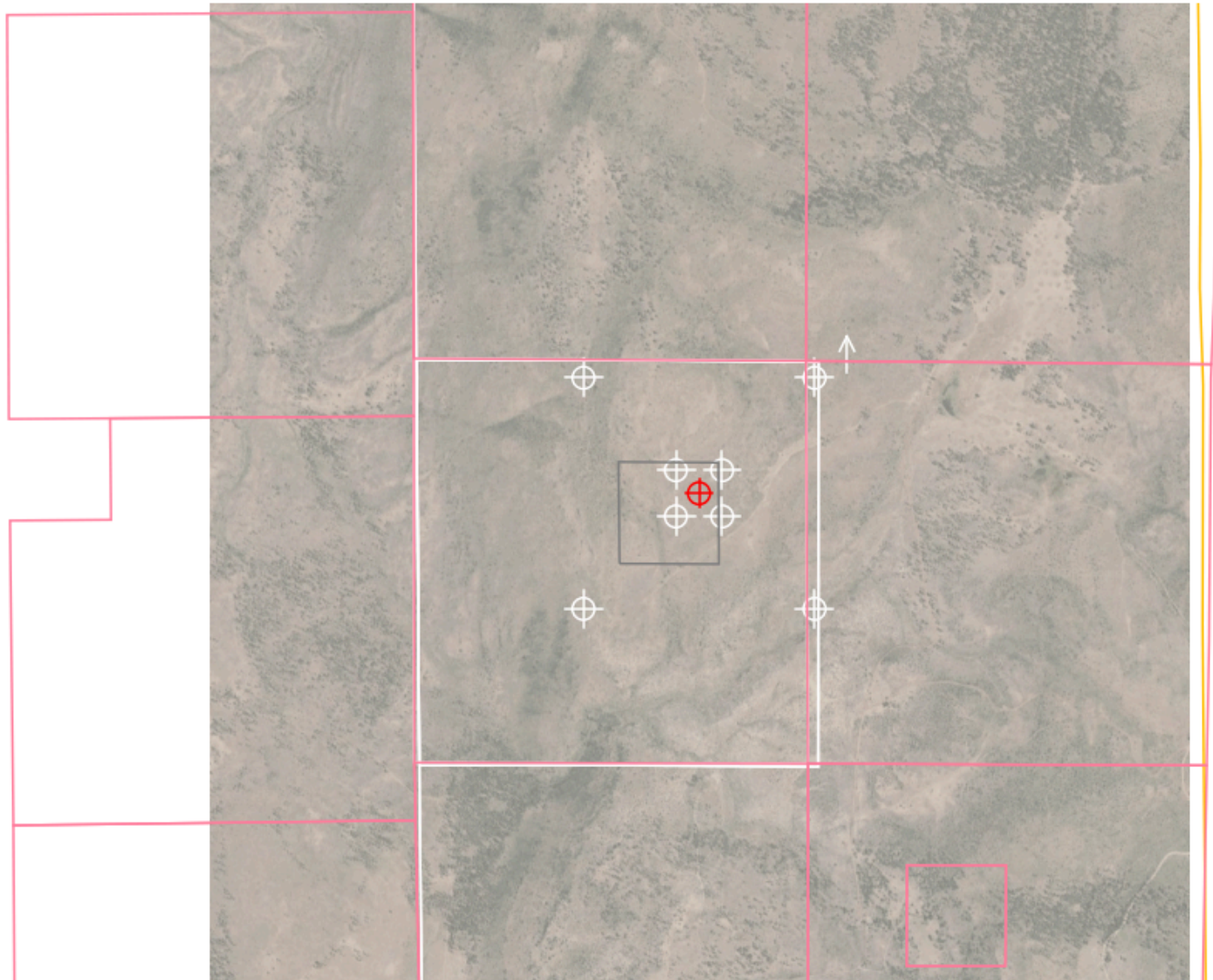


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Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.



TOPOSCOUT PARCEL OVERVIEW



Wide-area view of the subject parcel and adjoining parcels over aerial imagery. See the SITE DXF PREVIEW page for detailed survey layers.

TOPOSCOUT SITE PLAN PREVIEW



INCLUDED LAYERS



0 — 400 ft State Plane · NAD83 · US Feet

■ Contours (Index/Major/Minor/1ft) ■ PLSS Section/Township/QQ ■ Roads Major/Local/Trail + ROW ■ Buildings ■ Hydro ■ Power Lines ■ Pipeline ■ Railroad ■ Benchmarks + Labels
■ County Boundary ■ OTLS Grid ■ Site Info ■ Elevation Labels ■ 3D Polylines (all geometry)



SITE AERIAL – 600 ft × 600 ft

40.87092°N, 121.34479°W · Google Aerial



AREA CONTEXT — 3000 ft × 3000 ft

40.87092°N, 121.34479°W · Google Aerial

✂ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – [https://www.ngs.noaa.gov/cgi-](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

[bin/datasheet.prl](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – <https://geodesy.noaa.gov/GEOID/>

GEOID18 and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

